1-Bit Fluids

Lossless Compression for Deterministic Checkpointing of 3D Fluid Grids

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Video

Full paper

Viscous CFD simulations are spatially smooth, enabling compression. Quantization improves entropy encoding, and subtractive dither reduces quantization artifacts. But naively implemented, subtractive dither would balloon checkpoints by **71%**. This work uses ECDQ to reduce the penalty to **9%**, resulting in checkpoints with under **1 bit per component**, after lossless compression.

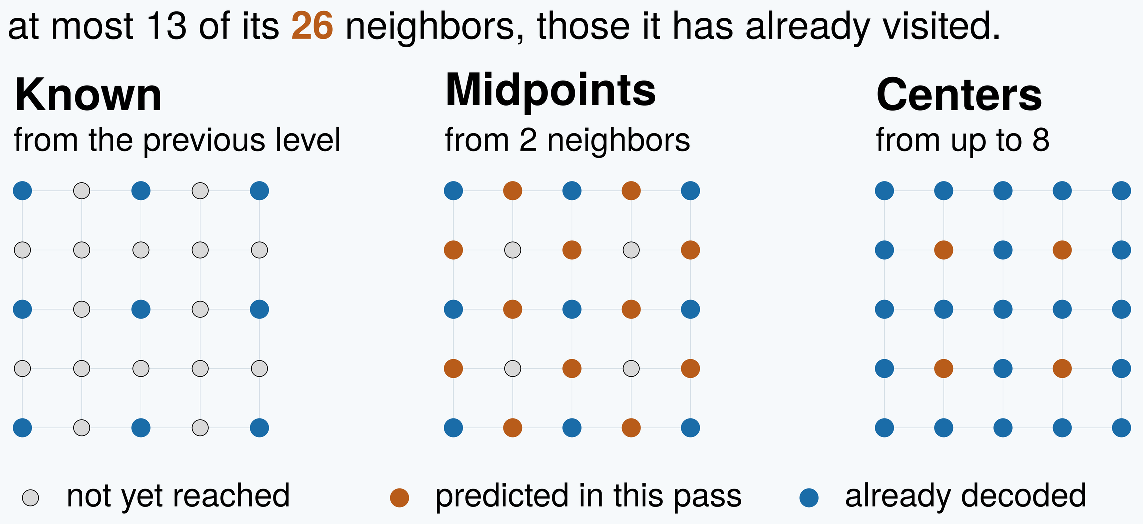
Deterministic Checkpointing

Deterministic checkpointing supports debugging during development, reproducible scientific investigation of results, mitigation of hardware failures and power outages, and flexible execution planning, including partial runs and runs that migrate across hardware. The computational fluid dynamics (CFD) velocity field dominates checkpoint size: at 256^3 with three fp16 components it would be **100 MB** by itself. For simulations running millions of timesteps, this storage cost becomes prohibitive.

The companion paper's *8-Bit Fluids* solver quantizes its checkpoint tap-off to dithered int8, halving this to 50 MB. The question addressed here is how far *lossless* compression can reduce these quantized fields, given that typical velocity fields in artificial life simulations are sparse: most of the fluid volume has smoothly varying velocity, with rich flow structure only around around motile cells.

Hierarchical Non-Causal Prediction

General-purpose predictive codecs scan in raster order because they were designed for streaming, but checkpoints are not streamed. They are 3D data structures decoded whole and offline. Prediction proceeds through a hierarchy of levels, following the lifting scheme, from stride $N/2$ down to stride 1: eight levels on a 256^3 grid. At each level new samples are predicted from their already-known neighbors, and the prediction residual is what gets encoded. The structure is non-causal, so each is based on its neighbors in all directions rather than only those that precede it in raster-scan order. A causal raster-scan predictor in 3D can use at most 13 of its **26** neighbors, those it has already visited.



One level of the hierarchy, shown as a 2D slice. New samples are processed in sub-grid passes of increasing density, each using the samples reconstructed in previous passes to make its predictions. In 3D a final body-center pass follows, predicted from all 26 surrounding neighbors with weights inversely proportional to distance.

> **The predictor also functions as a noise filter.**
>
> The velocity field has a noise floor corresponding to int8 quantization, and dither makes that error independent from sample to sample. In the body-center pass, inverse-distance weighting over 26 neighbors cuts it to roughly **0.20** of its unfiltered value, barely worse than the $1/\sqrt{26}$ a uniform average would give, while a causal predictor limited to 13 neighbors manages only about 0.28, worse by exactly $\sqrt{2}$. The underlying velocity field is usually smooth enough that even the corner-neighbors introduce little bias. The predictor therefore **sees through** the noise to recover the underlying signal.

The solver's periodic boundary conditions wrap the grid into a 3D torus, eliminating edge effects so samples near edges have the same neighborhood structure as interior samples.

Divergence-Free Constraint

Since the velocity components u_x and u_y are decoded before u_z, the incompressibility constraint $\nabla \cdot \mathbf{u} = 0$ can in principle supply additional predictive information for u_z: given the spatial derivatives of the already-decoded components, the divergence-free condition constrains what $\partial u_z/\partial z$ must be.

Divergence is calculated using finite differences, which amplify high-frequency noise. Therefore, in practice the benefit is small, **0.4%** at 8-bit quantization and roughly 1% at 10 bits. At practical quantization levels, spatial correlation between neighboring samples swamps divergence as a source of predictive information.

Dither Synchronization

Subtractive dither removes signal-correlated quantization distortion. If the encoder naively predicts from dithered quantized values, however, the random offsets effectively randomize the rounding thresholds between neighbors: two samples with identical true values may round differently because they received different dither values, producing a nonzero residual where a zero would otherwise result. With 8-bit quantization this inflated the encoded size from **0.882 to 1.507 bps**, a 71% increase.

That increase is unnecessary: the dither is reproducible from the same seed on both sides, so it carries no information and in principle costs nothing. What it actually costs is a **rounding-threshold mismatch**: a sample is quantized via its own jittered threshold, and a prediction built from its neighbors includes their jittered thresholds. Removing the mismatch removes the cost. The predictor operates on de-dithered values, and the sample's own dither is injected before the prediction is rounded. Both are then rounded using the same threshold, so an accurate prediction results in a residual of exactly zero, no matter what dither value was drawn. **The penalty falls from over 70% to under 9%.**

> **Dither-synchronized predictive codec**
>
> Upstream quantization: $q_{n,t} = \mathrm{round}(v_{n,t} + d_{n,t})$
>
> **Encoder**
>
> **for each** sample n:
>
> $\hat{y}_j = q_{j,t} - d_{j,t}$
>
> $p = \frac{\sum_j w_j \hat{y}_j}{\sum_j w_j}$
>
> $p_q = \mathrm{round}(p + d_{n,t})$
>
> $r_n = q_{n,t} - p_q$
>
> emit r_n
>
> **Decoder**
>
> **for each** sample n:
>
> $\hat{y}_j = q_{j,t} - d_{j,t}$
>
> $p = \frac{\sum_j w_j \hat{y}_j}{\sum_j w_j}$
>
> $p_q = \mathrm{round}(p + d_{n,t})$
>
> $q_{n,t} = r_n + p_q$
>
> n: current sample. j: neighbor indices. t: timestep. v: velocity. q: quantized dithered velocity. r: residual. d: reproducible dither $U(-0.5, +0.5)$. w_j: distance weights. Unit quantization step assumed. At the coarsest level no neighbors exist, the sum over j is empty, and $p = 0$.

Encoder and decoder are identical except for the last step: subtract to produce a residual, add to reconstruct.

The Quiet Zone and ECDQ

Without dither, neighboring near-zero values all quantize to exactly zero, producing long runs of zeros that compress very efficiently. Dither disrupts these uniform regions (the quiet zone) by randomizing near-zero values to ± 1.

As resolution increases the quiet zone shrinks, from **36% of samples at 8 bits to 9% at 10 bits**, the dither entropy penalty diminishes, and the entropy-constrained dithered lattice quantization (**ECDQ**) prediction quality improvement begins to dominate.

> **This illuminates a tradeoff between two phenomena pointing in opposite directions.**
>
> - **Schobben et al.** characterized the entropy penalty of dithered quantization: dither increases the information content of a quantized signal.
> - **Zamir and Feder** showed that an entropy-coded dithered quantizer inside a prediction loop can achieve rate-distortion optimality. This is where ECDQ comes from.
> - Neither source investigated where the two phenomena crossover on real data. **Here the crossover is at roughly 9 bits.**

Split3 Encoding (Three-Stream)

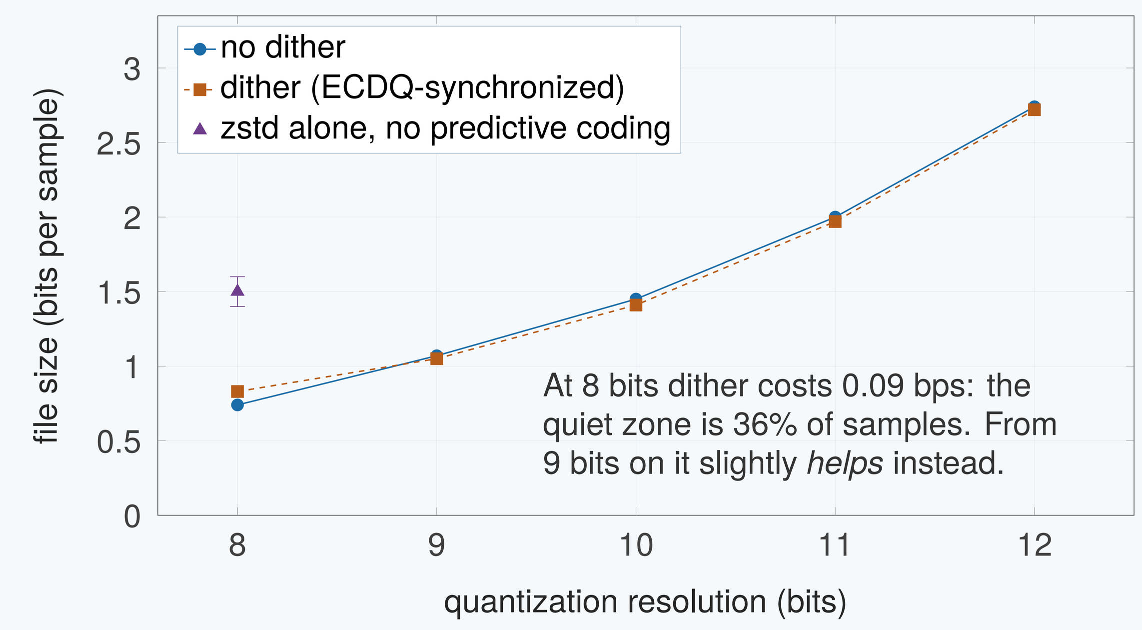
Prediction residuals are split into three byte-streams, each compressed independently with zstd: 1) a bitpacked zero/nonzero bitmap, 2) bitpacked sign bits for the nonzeros only, and 3) their magnitudes minus one as variable-length integers. Splitting into **homogeneous streams** gives zstd's LZ77 backend more regular byte patterns than a mixed representation, and outperforms zigzag-plus-varint by about 8%.

Results

All results were verified to be bit-exact by writing compressed files to disk and decompressing them, then comparing before and after. The data comes from 256^3 velocity fields, extracted from an artificial life simulation containing 1000 motile cells with diameters of 4 to 10 lattice units.

	File size (bps)		Entropy H (bps)		RMS
Bits	No dither	Dither	No dither	Dither	(lu/step)
8	0.74	0.83	0.76	0.83	0.00226
9	1.07	1.05	1.09	1.06	0.00112
10	1.45	1.41	1.50	1.45	0.00056
11	2.00	1.97	2.06	2.03	0.00028
12	2.74	2.72	2.80	2.79	0.00014

File size and Shannon entropy H in bits per sample, averaged across velocity components. H is i.i.d. symbol entropy and not a lower bound: zstd also exploits correlation between nearby symbols, so file sizes land up to 3% below it. The gap is small because the predictor has already removed most of that structure. RMS in lattice units per timestep.



Predictive coding pays off: general-purpose compressors applied directly to the same 8-bit quantized data achieve only 1.4 to 1.6 bps. The curves cross just shy of 9 bits.

- **8 bits, no dither:** 0.743 bps, **4.7 MB** checkpoints.
- **8 bits, with dither:** 0.831 bps, **5.2 MB** checkpoints. (Both are well under 1 bit per sample.)
- **Graceful degradation on adversarial data.** As a stress test, a large unphysical velocity source producing flow speeds far exceeding typical speeds in alife simulations resulted in approximately 2.3 bps at 8 bit quantization.

Noise Floor Amplitude Versus Rate

The two rightmost quantities in the results table can be interpreted together. RMS is the amplitude of the noise floor the companion solver provides to the encoder; bits per sample is the cost.

> Halving the amplitude of the noise floor costs approximately **half a bit per sample**. From 8 to 12 bits, RMS falls by a factor of 16 while the rate rises from 0.74 to 2.74 bps.

100 MB → 6.3 MB

For fluid-grid checkpointing with a 256^3 grid running at 190 timesteps per second on an RTX 3090, even aggressive checkpointing every few seconds of simulation time produces manageable storage requirements. This makes regular deterministic checkpointing of fluid-mediated evolutionary runs spanning millions of timesteps practical on a single consumer GPU with consumer-level storage devices.

While developed for an artificial life context, the individual components of this scheme are **general-purpose**. The hierarchical non-causal predictor, ECDQ-style dither synchronization, and split3 stream encoding are applicable to any spectral fluid solver with periodic boundaries that produces sparse velocity fields.

References

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 • Video: youtu.be/3dRSOxKgSgg • Full paper: doi.org/10.5281/zenodo.21844937

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